LEXI Angles

Michael A. Shoemaker[1*]

[1]Navigation and Mission Design Branch, NASA Goddard Space Flight Center, 8800 Greenbelt Rd., Greenbelt, 20771, MD, USA.

Corresponding author(s). E-mail(s): michael.a.shoemaker@nasa.gov;

1 Introduction

Let $\mathscr{F}_b$ be the reference frame fixed to the lander body, and $\mathscr{F}_d$ be the reference frame fixed to the LEXI detector. At any point in time, we wish to write the rotation matrix $\mathbf{R}^{db}$:

$$\mathbf{v}_d = \mathbf{R}^{db}\mathbf{v}_b \tag{1}$$

that maps an arbitrary vector $\mathbf{v}_b$ expressed in $\mathscr{F}_b$ into the corresponding vector $\mathbf{v}_d$ expressed in $\mathscr{F}_b$. Note that $\mathbf{R}^{bd} = \mathbf{R}^{db^T}$. The matrix $\mathbf{R}^{db}$ is a function of three variables:

$$\mathbf{R}^{db} = f(\theta_1(t), \theta_2(t), \theta_3) \tag{2}$$

The angles θ_1 and θ_2 are explicitly time-dependent, and can be thought of as either the Az-El or the RA-DEC angles, but that distinction is not important for the present discussion. The angles θ_1 and θ_2 define the direction of the detector boresight, and the angle θ_3 defines the "roll" of the detector's horizontal and vertical axes about the boresight. Note that θ_3 does not vary in time, because the gimbal has no actuation about the boresight; the value of θ_3 is determined by the mechanical design of the detector plate relative to the gimbal. The purpose of this memo is to describe how to measure θ_3 from the LEXI CAD model.

The reference frames are defined by their unit basis vectors, $\mathscr{F}_b = \{\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3\}$ and $\mathscr{F}_d = \{\hat{\mathbf{d}}_1, \hat{\mathbf{d}}_2, \hat{\mathbf{d}}_3\}$. The matrix $\mathbf{R}^{db}$ is a Direction Cosine Matrix (DCM), which means that each element R_{ij} of $\mathbf{R}^{db}$ is the direction cosine between $\hat{\mathbf{d}}_i$ and $\hat{\mathbf{b}}_j$. Thus, the first step is to use CAD software to measure the angle between the unit basis vectors. As shown in Fig. 1, the frame $\mathscr{F}_b$ is defined using the same convention that FireFly uses for the Lander Body Frame, where $\hat{\mathbf{b}}_1$ points towards zenith out the top deck, $\hat{\mathbf{b}}_2$ points towards north during a nominal landing, and $\hat{\mathbf{b}}_3$ points west during a nominal landing. The detector frame $\mathscr{F}_d$ is defined with $\hat{\mathbf{d}}_1$ and $\hat{\mathbf{d}}_2$ aligned with Ramiz's convention (X goes positive to the right in an image, Y goes positive down in an image), and $\hat{\mathbf{d}}_3$ is parallel to the boresight with positive direction going from detector to observation target. Based on the measured angles within CAD, the DCM

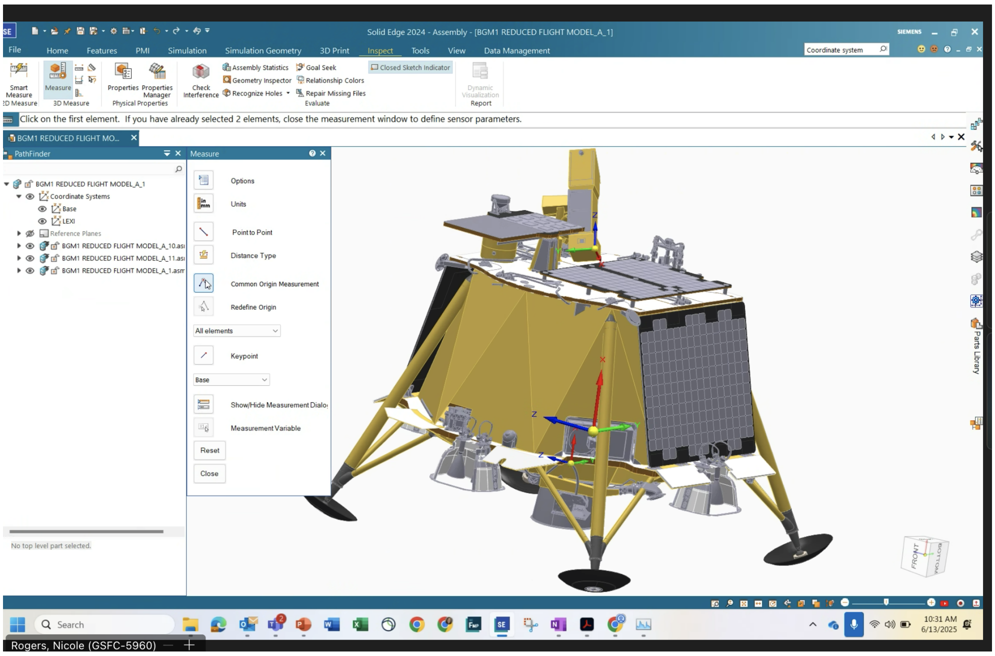

Fig. 1 LEXI and BGM1 CAD models with detector frame and lander body frames defined.

of $\mathbf{R}^{db}$ was found to be:

$$\mathbf{R}^{db} = \begin{bmatrix} \cos(180 - 55.36) & \cos(38.17) & \cos(75.96) \\ \cos(180 - 76.31) & \cos(116.02) & \cos(29.89) \\ \cos(180 - 142.0) & \cos(64.19) & \cos(64.19) \end{bmatrix} \tag{3}$$

$$= \begin{bmatrix} -0.56841826 & 0.78618058 & 0.24259923 \\ -0.23666858 & -0.43868486 & 0.86698374 \\ 0.78801075 & 0.43538823 & 0.43538823 \end{bmatrix} \tag{4}$$

where all the angles above are in degrees, and the terms having the 180 inside the cosine were to account for the correct sign, since the various CAD programs were inconsistent in how they measured angles between vectors. Note that another interpretation of the DCM is as a collection of column vectors of one set of basis vectors expressed in the other frame, namely:

$$\mathbf{R}^{db} = [\hat{\mathbf{b}}_1^d \ \hat{\mathbf{b}}_2^d \ \hat{\mathbf{b}}_3^d] \tag{5}$$

where the superscript on a basis vector denotes the frame $\mathscr{F}$ in which it is expressed. The understanding in Eq. 5 allowed us to check the sign of the individual terms R_{ij}

to decide whether the 180 needed to be added to Eq. 3. The screenshots below give convenient visual verification of the sign needed in the components in Eq. 5.

We can double check that the numerical values given in Eq. 3 represent a valid rotation matrix in a couple of ways. One property of a rotation matrix is that the norm of the columns and rows should equal 1. Performing this calculation results in

$$\text{Norms along rows} = \begin{bmatrix}1.00001681 & 1.00005861 & 1.00004338\end{bmatrix} \tag{6}$$

$$\text{Norms along columns} = \begin{bmatrix}1.00003614 & 1.00004361 & 1.00003905\end{bmatrix} \tag{7}$$

which are sufficiently close to 1. Another check is that any rotation matrix is orthogonal, and thus $\mathbf{R}^{-1} = \mathbf{R}^T$, which we can check with

$$\mathbf{R}^{-1} - \mathbf{R}^T = \begin{bmatrix} \text{1.30566263e-05} & \text{2.77672763e-05} & \text{-7.39737298e-05} \\ \text{-3.87599546e-05} & \text{8.35426602e-05} & \text{-4.61712640e-05} \\ \text{1.76195339e-05} & \text{-8.53548825e-05} & \text{-1.92275805e-05} \end{bmatrix} \tag{8}$$

which is sufficiently close to zero. We therefore can have confidence that $\mathbf{R}^{db}$ given above is a valid rotation matrix.

We can also visually check that $\mathbf{R}^{db}$ is correct by doing a visual inspection of the $\mathscr{F}_d$ basis vectors after being transformed with $\mathbf{R}^{bd}$ into $\mathscr{F}_b$ and comparing the illustrations of the frames in the CAD drawings. In Figs. 5 to 7 below, the 3D plot on the left shows the predicted $\mathscr{F}_d$ relative to $\mathscr{F}_b$ given the $\mathbf{R}^{bd}$ computed in Eq. 3, and the right half of each figure shows the CAD view of the lander in the matching orientation; visual inspection allows us to confirm (within the accuracy allowed by this sanity check) that the frames are oriented in the proper direction.

The next step is to find the roll angle given $\mathbf{R}^{db}$. Let the sequence of rotations implied by Eq. 2 be given by the intrinsic 1-2-3 Euler angle sequence:

$$\mathbf{R}^{db} = \mathbf{R}_3(\theta_3)\mathbf{R}_2(\theta_2)\mathbf{R}_1(\theta_1) \tag{9}$$

where θ_3 is the roll angle we wish to find. The equation for this 1-2-3 sequence is

$$\mathbf{R}^{bd} = \begin{bmatrix} c_3c_2 & c_3s_2s_1 + s_3c_1 & -c_3s_2c_1 + s_3s_1 \\ -s_3c_2 & -s_3s_2s_1 + c_3c_1 & s_3s_2c_1 + c_3s_1 \\ s_2 & -c_2s_1 & c_2c_1 \end{bmatrix} \tag{10}$$

where $c_1 = \cos\theta_1$, $s_1 = \sin\theta_1$, etc. We can solve for θ_3 using

$$\theta_3 = \text{atan2}\left(-R_{2,1}, R_{1,1}\right) \tag{11}$$

which, given the numerical values for $\mathbf{R}^{db}$ above, give $\theta_3 = 157.3949$ degrees. We can also solve for θ_1 and θ_2 if we want, but there isn't much need for them (besides visually checking the illustrations above) because at any point in time these angles will instead come from the actual boresight direction.

Note that a few more steps are required to compute the correct $\mathbf{R}^{db}$ given the azimuth and elevation angles measured at a given time. That works out to:

$$\mathbf{R}^{db} = \mathbf{R}_3(\theta_3)\mathbf{R}_2(\theta_2)\mathbf{R}_1(-\theta_1)\mathbf{R}_1(-90) \tag{12}$$

where θ_3 is the value given just above, θ_1 is the azimuth angle measured clockwise from north, θ_2 is the elevation angle measured positive from the horizontal plane.

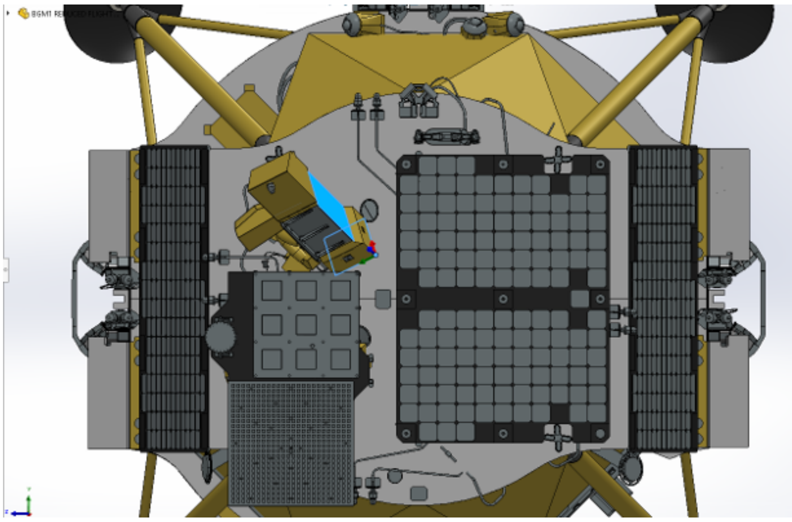

Fig. 2 View of Lander Y-Z frame.

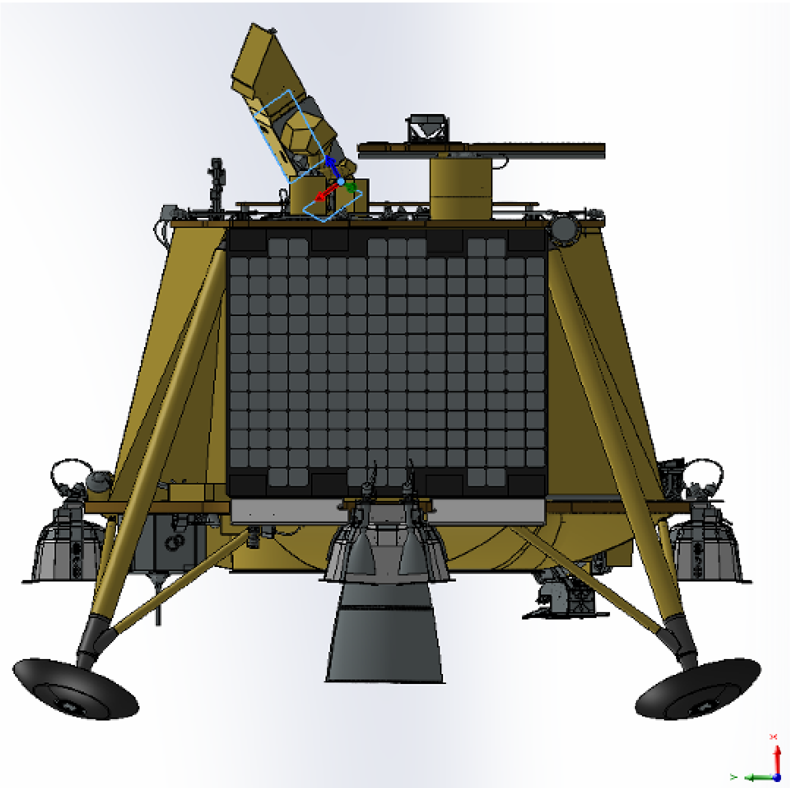

Fig. 3 View of Lander X-Y frame.

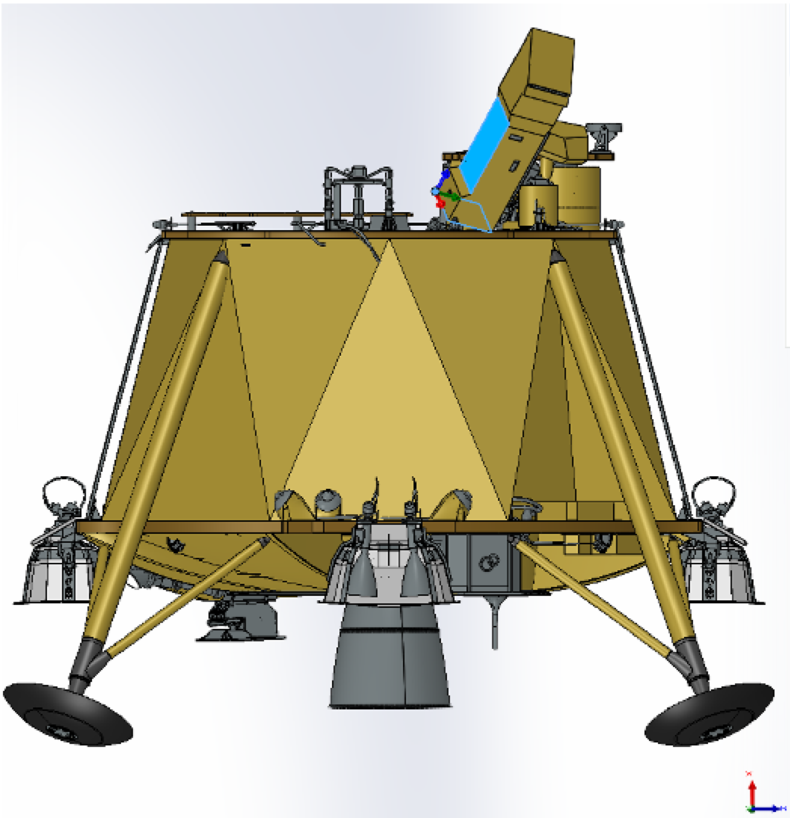

Fig. 4 View of Lander X-Z frame.

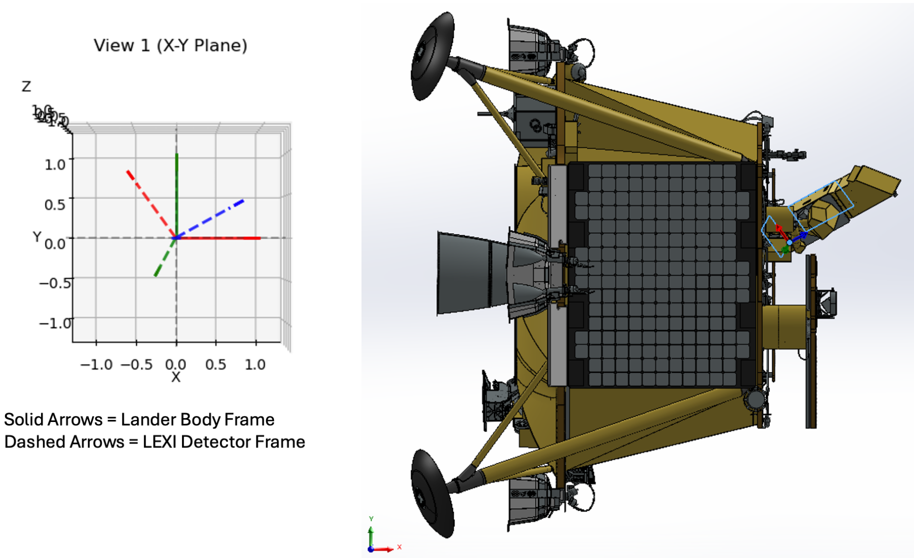

Fig. 5 View of Lander X-Y frame with check of rotated basis vectors.

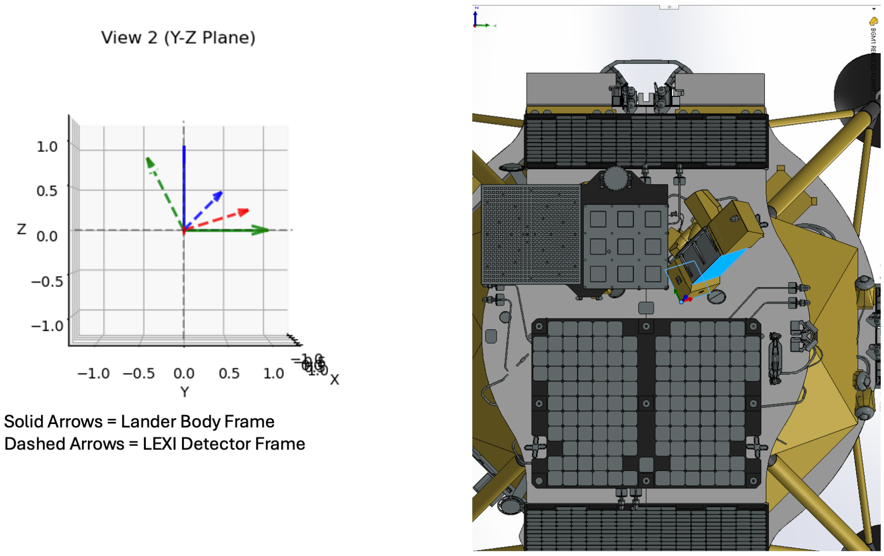

Fig. 6 View of Lander Y-Z frame with check of rotated basis vectors.

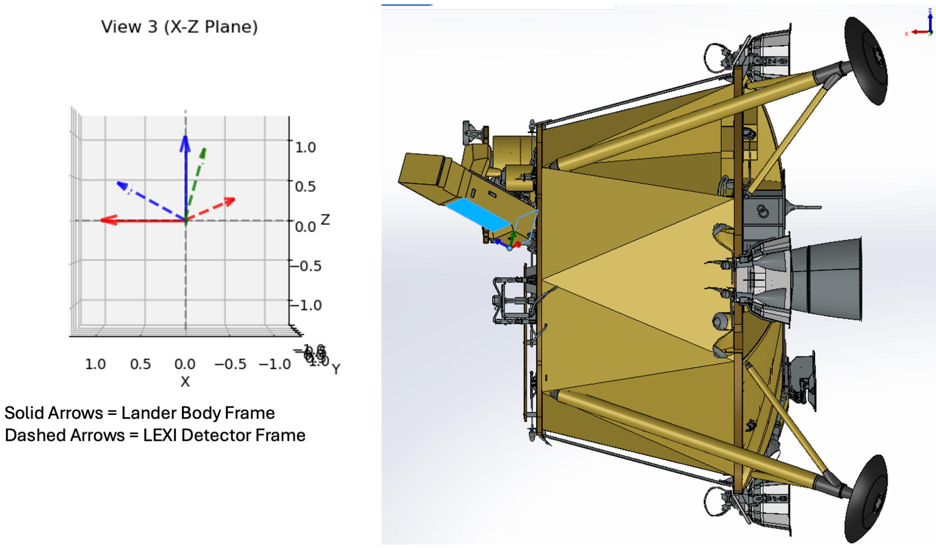

Fig. 7 View of Lander X-Z frame with check of rotated basis vectors.